

# EnergiSmart — Reference Datasets

All hardcoded data for data.py, with sources for each. Compiled 25 August 2026.

## 1. TNB Domestic Tariff (effective 1 July 2025)

| Charge Type                             | Rate                                           |
|-----------------------------------------|------------------------------------------------|
| Energy Charge ( $\leq 1,500$ kWh/month) | 27.03 sen/kWh                                  |
| Energy Charge ( $> 1,500$ kWh/month)    | 37.03 sen/kWh                                  |
| Capacity Charge                         | 4.55 sen/kWh                                   |
| Network Charge                          | 12.85 sen/kWh                                  |
| Retail Charge                           | RM10.00/month (waived if usage $\leq 600$ kWh) |

**Source:** Tenaga Nasional Berhad (TNB), official Domestic Tariff Schedule, effective 1 July 2025 — [mytnb.com.my/tariff/index.html](https://mytnb.com.my/tariff/index.html). (Already in the user's own FYP notes.)

## 2. Energy Efficiency Incentive (Rebate Table)

Rebate rate depends on total monthly usage tier (sen/kWh):

| Usage (kWh/mo) | Rebate (sen/kWh) | Usage (kWh/mo) | Rebate (sen/kWh) |
|----------------|------------------|----------------|------------------|
| 1–200          | 25.0             | 601–650        | 7.5              |
| 201–250        | 24.5             | 651–700        | 5.5              |
| 251–300        | 22.5             | 701–750        | 4.5              |
| 301–350        | 21.0             | 751–800        | 4.0              |
| 351–400        | 17.0             | 801–850        | 2.5              |
| 401–450        | 14.5             | 851–900        | 1.0              |
| 451–500        | 12.0             | 901–1000       | 0.5              |
| 501–550        | 10.5             | Above 1000     | No rebate        |
| 551–600        | 9.0              |                |                  |

**Source:** Same TNB Domestic Tariff Schedule as above — [mytnb.com.my/tariff/index.html](https://mytnb.com.my/tariff/index.html). (Already in the user's own FYP notes.)

## 3. Appliance Wattage — Official 10-Appliance List

| # | Appliance       | Wattage                        |
|---|-----------------|--------------------------------|
| 1 | Air Conditioner | Per HP table (Section 4 below) |
| 2 | Refrigerator    | 400 W                          |
| 3 | Washing Machine | 500 W                          |
| 4 | Dryer           | 3,400 W                        |
| 5 | Water Heater    | 1,000 W                        |
| 6 | Fan             | 50 W                           |
| 7 | Television      | 50 W                           |
| 8 | Desktop PC      | 300 W                          |

| #  | Appliance           | Wattage |
|----|---------------------|---------|
| 9  | Laptop              | 50 W    |
| 10 | Lighting (per bulb) | 32 W    |

**Source:** TNB Home Energy Calculator — [hec.tnb.com.my/Public/UsageCalculator](https://hec.tnb.com.my/Public/UsageCalculator). (Already in the user's own FYP notes.) Note: build strictly to this table, not the prototype mockup's appliance list (rice cooker/ceiling fan not used).

#### 4. Air Conditioner: Horsepower (HP) → Watt Conversion

| HP     | Watt    |
|--------|---------|
| 0.5 HP | 373 W   |
| 1.0 HP | 746 W   |
| 1.5 HP | 1,119 W |
| 2.0 HP | 1,492 W |
| 2.5 HP | 1,865 W |
| 3.0 HP | 2,238 W |

**Source:** TNB Home Energy Calculator — [hec.tnb.com.my/Public/UsageCalculator](https://hec.tnb.com.my/Public/UsageCalculator). (Already in the user's own FYP notes.)

#### 5. CO<sub>2</sub> Emissions Factor

| Year                        | GEF — Peninsular Malaysia (kg CO <sub>2</sub> /kWh) |
|-----------------------------|-----------------------------------------------------|
| 2022                        | 0.769                                               |
| 2023                        | 0.760                                               |
| 2024 (used in this project) | 0.740                                               |

**Source:** Suruhanjaya Tenaga (Energy Commission Malaysia), "Grid Emission Factor (GEF) in Malaysia, 2022–2024 (Provisional)" — [myenergystats.st.gov.my/documents/d/guest/grid-emission-factor-gef-in-malaysia-2022-2024-provisional-](https://myenergystats.st.gov.my/documents/d/guest/grid-emission-factor-gef-in-malaysia-2022-2024-provisional-). NOT from the user's notes — the FYP report (p.8, p.22) names "Suruhanjaya Tenaga (2024)" as the source but never states the figure; this number was retrieved from that official document directly. User should personally verify it before citing it in a viva.